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Sourcing List — Automated Test Station

> 📄 **Print-ready PDF:** `docs/pdf/sourcing-list.pdf`

Shopping list for the breadboard prototype and first full build. Grouped by priority — Priority 1 blocks breadboarding and must be ordered before any bench work can start.

Legend: ✅ on hand · ⏳ ordered · 🛒 to source

Approximate prices are USD, mid-2026 retail. Use as ballparks, not quotes.

Priority 1 — Order Today (blocks breadboarding)

These parts are required before you can do anything on the breadboard. Order all of them in one cart.

#	Item	Qty	Status	Approx $	Notes
1	MAX31855 thermocouple amplifier breakout	3	🛒	~$5 ea	Amazon search: **"MAX31855 thermocouple amplifier breakout"** — Adafruit #269 or compatible clone. Exposes VCC, GND, DO, CS, CLK on a 5-pin SIP header. Verify breakout has 3.3V regulator or is marked 3.3V-only — the MAX31855 is NOT 5V.
2	Solderless breadboard, 830-point, full-size	1	🛒	~$8	Only if not already on hand. Amazon: **"830 point solderless breadboard"**. A half-size (400-point) will be too tight for 3 modules + protection resistors — get full-size.
3	E-stop mushroom button, NC, latching, 22mm panel mount	1	🛒	~$6	Amazon: **"22mm latching mushroom emergency stop button NC"** — red head, normally-closed contacts. Must be latching (twist-to-release), not momentary. This is a safety-critical component — don't use the cheapest listing; pick one with 10A/250VAC contact rating.
4	Green momentary pushbutton, NC, 22mm panel mount	1	🛒	~$3	Amazon: **"22mm momentary pushbutton green NC normally closed"** — manual START button wired to LOGO! I4. NC type allows fail-safe wiring.
5	Red momentary pushbutton, NC, 22mm panel mount	1	🛒	~$3	Amazon: **"22mm momentary pushbutton red NC normally closed"** — manual STOP button wired to LOGO! I5.

Priority 1 subtotal: ~$25–30

Priority 2 — Order Soon (needed for heater circuit)

Order these alongside Priority 1 or immediately after. The heater circuit cannot be tested or brought up without them.

#	Item	Qty	Status	Approx $	Notes
6	Fotek SSR-40DA solid-state relay	1	🛒	~$8–12	Amazon: **"SSR-40DA solid state relay"**. Spec: 3–32V DC control input, 24–480V AC load side, 40A, zero-crossing. At 150W/120VAC the load is 1.25A — well within rating. Must be genuine Fotek or a quality clone; cheap counterfeits have caused fires. Check seller reviews carefully.
7	150W 120VAC cartridge heater	1	🛒	~$12–18	Amazon: **"cartridge heater 150W 120V"** or search Omega part **"CIR-1012/120V"**. Common size: 5/16" (8mm) diameter × 3" (76mm) long. Verify 120VAC specifically — 240VAC units exist and look identical.
8	KSD301 bimetallic thermostat disc, 120°C, NC	2	🛒	~$1–2 ea	Amazon: **"KSD301 thermostat 120C NC normally closed"** — snap-disc type, mounts directly against heater body. 120°C trip point = hardware overtemp cutout wired to LOGO! I3. Buy 2: one installed, one spare.
9	Thermal cutoff fuse, 150°C, 10A, axial	5	🛒	~$1–2 ea	DigiKey/Mouser: search **"thermal cutoff fuse 150C 10A axial"**. Or Amazon: **"152C thermal fuse 10A"** — 152°C is a common standard rating close to 150°C. Spec: one-time non-resettable TCO type. Must be rated ≥10A. Clip directly to heater body — NOT in free air. Buy 5: 1 installed + 4 spares (a blown fuse means investigate; don't reuse).
10	Aluminium heatsink for SSR	1	🛒	~$5–8	Amazon: **"SSR heatsink aluminum solid state relay"** — look for a dedicated SSR heatsink with pre-drilled mounting holes matching 40A SSR footprint (57.5mm × 44mm typical). Without a heatsink the SSR overheats at sustained duty. A DIN-rail mounted extrusion with fins ≥50 cm² face area also works.

Priority 2 subtotal: ~$35–50

Priority 3 — Servo Vent Control

Order when you are ready to wire the vent servos. Not needed for initial heater bring-up.

#	Item	Qty	Status	Approx $	Notes
11	Tower Pro MG90S metal-gear servo (or SG90 plastic)	2	🛒	~$3–5 ea	Amazon: **"MG90S metal gear servo"** — metal gear recommended for repeated vent duty. SG90 plastic gear is fine for light loads. Standard 50Hz PWM, 500–2500µs pulse, 5V supply.
12	5V 2A BEC / DC-DC converter, 24V input	1	🛒	~$5	Amazon: **"24V to 5V DC DC converter 2A step down"** — powers servo rail from LOGO! Q2 24V relay output. Needs to handle 1A peak (2 servos at stall). Get a switching BEC, not a linear regulator (24→5V linear = 19V × 1A = 19W of heat = not good).

Priority 3 subtotal: ~$12–20

Already Have (verify before ordering)

Check your bench stock before adding these to a cart.

#	Item	Qty	Status	Notes
13	K-type thermocouples with leads	3	✅	One per MAX31855 channel — TC1 ambient, TC2 DUT, TC3 heater. Yellow wire = positive (ANSI standard).
14	Dupont jumper wires (M-F, F-F assortment)	—	✅	For breadboard-to-RPi header connections.
15	Siemens LOGO! 12/24RCE PLC	1	✅	Already installed / on hand.
16	24V DIN rail PSU (Meanwell HDR-60-24 or equivalent)	1	✅	Powers LOGO! and relay coils.
17	WeMos D1 Mini / NodeMCU ESP8266	1	✅	Used for status-display sketch. NodeMCU (27mm row-spacing) will work on a breadboard; D1 Mini preferred for clean mounting.
18	Arduino resistor kit — 100Ω and 10kΩ	—	✅	Probably — confirm you have at least 3× 10kΩ (CS pull-ups) and 3× 100Ω (protection resistors). If stock is low, order a 1/4W resistor assortment on Amazon: **"resistor assortment kit 1/4W 600 piece"**.
19	100nF (0.1µF) ceramic capacitors	3+	✅	Probably — confirm at least 3× 100nF for MAX31855 VCC decoupling.
20	SSD1306 128×64 OLED display (I²C)	1	✅	Check Arduino kit. If not on hand: Amazon **"SSD1306 0.96 inch OLED I2C"** ~$4.
21	Raspberry Pi 4B	1	❓	**Required for GPIO.** The Flask app can run on the NUC for development, but reading MAX31855 via SPI and driving GPIO12/13/18 for SSR and servos requires RPi GPIO. If you don't have one, add a RPi 4B (2GB) to Priority 1 (~$35–45, Adafruit or PiShop — avoid grey-market pricing). The RPi 4B must have SPI enabled (`raspi-config → Interface Options → SPI → Enable`).

Note on Raspberry Pi

The thermal controller as designed requires an **RPi 4B** for GPIO access: - SPI0 hardware bus (GPIO8/7/25 for CS, GPIO11/9 for CLK/MISO) → reads MAX31855 modules - GPIO12 hardware PWM → heater SSR duty cycle control - GPIO13, GPIO18 hardware PWM → servo A and servo B vent control

The **Flask HMI app** (`app.py`) can run on any host (NUC, laptop) during development with the hardware-dependent routes stubbed out. But for full rig operation with live temperature feedback and heater control, you need the RPi on the bench.

If you don't already have a Pi 4B, add it to the Priority 1 order. The 2GB RAM model is sufficient. Adafruit and PiShop.us are the most reliable US retailers; avoid Amazon third-party listings that may be overpriced or grey-market.

Total Cost Estimate

Priority bucket	Subtotal
Priority 1 — breadboard unblock	~$25–30
Priority 2 — heater circuit	~$35–50
Priority 3 — servos	~$12–20
RPi 4B (if needed)	~$35–45
Total to source	**~$110–145**

Minimum spend to get a breadboard bench test running (Priority 1 only, assuming RPi on hand): **~$25–30**.